

HAT-P-19b

R-1248

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_231017 · created 2026-07-31T11:38:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460234.8013 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.024 +/- 0.053
Transit depth (Rp/Rs)^2	0.056 +/- 0.25 [%]
Orbital Inclination (inc)	84.9 +/- 1.59
Airmass coefficient 1 (a1)	1.48 +/- 0.87
Airmass coefficient 2 (a2)	-0.34 +/- 0.84
Scatter in the residuals of the lightcurve fit is	61.8 %
Best Comparison Star	None
Optimal Aperture	2.2
Optimal Annulus	7.34
Transit Duration (day)	0.02 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.024 ± 0.053 vs 0.1418 (deviation 2.03σ)
Duration fit vs published	0.02 d vs 0.1196 d (deviation 2.68σ)
Mid-time O–C	-4.0 min
Depth SNR	0.4
Residual scatter	61.8 %

Light curve

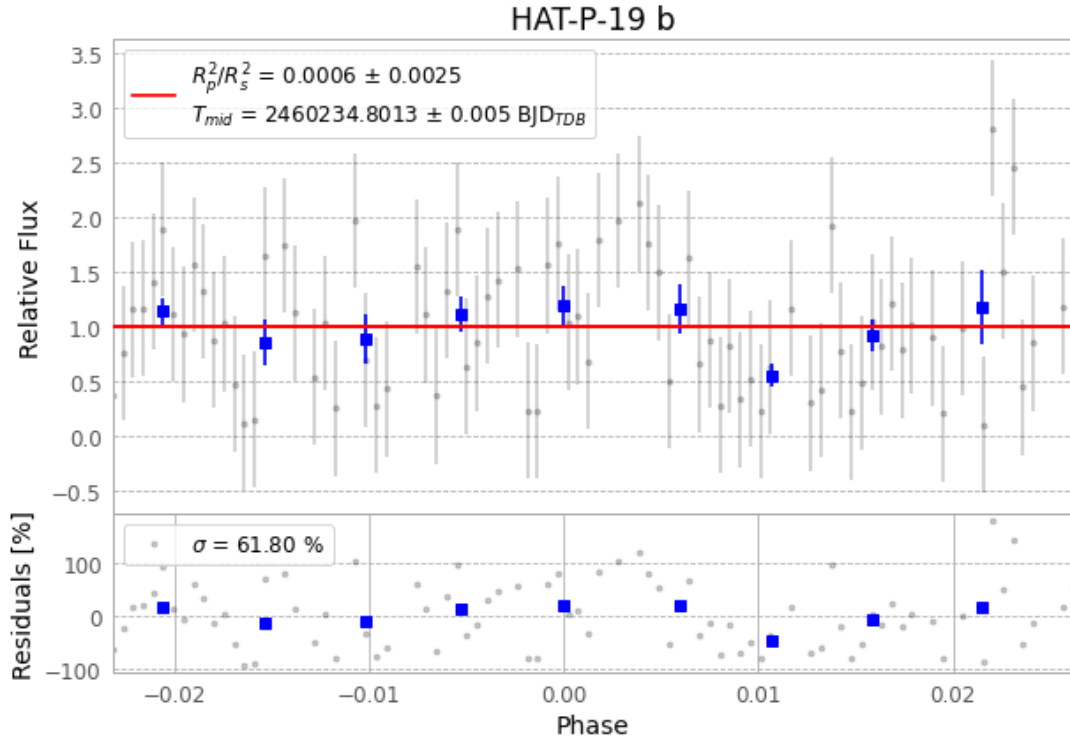


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-16.png (SHA-256 d806365ab02e7667...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [200, 434], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0662db1872f5ef6c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), HATP-19231017044815.FITS → HATP-19231017093616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231017.log (edae88c1b8f8...), FinalLightCurve_HAT-P-19 b_2023-10-16.png (d806365ab02e...), FinalLightCurve_HAT-P-19 b_2023-10-16.csv (9a8fb8d68681...)

Compute resources

Wall time 160.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:38Z.